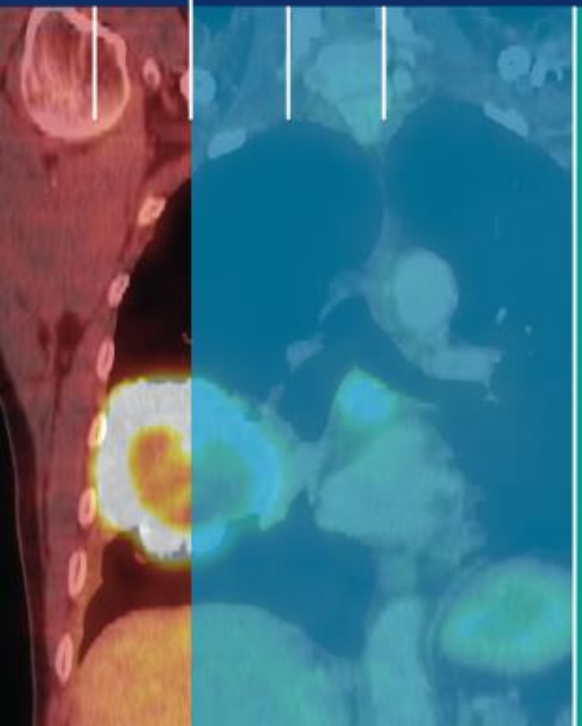
Knowledge Engineering in Radiation Oncology – A real-world application of Semantic Web technology



Andre Dekker, PhD

Medical Physicist

Information & Services Head

Knowledge Engineering Head

MAASTRO Clinic

Met de steun van het Europees Fonds voor Regionale Ontwikkeling.

De Europese Commissie investeert in uw toekomst.

Disclosures

- Research collaborations incl. funding
  - Varian (VATE, chinaCAT, euroCAT), Siemens (euroCAT), Sohard (SeDI), Philips, Xerox (EURECA)
- Public research funding
  - Radiomics (USA-NIH/U01CA143062), euroCAT(EU-Interreg), duCAT (NL-STW), EURECA (EU-FP7), SeDI (EU-EUREKA), TraIT (NL-CTMM), DLRA (NL-NVRO)
- Spin-offs and commercial ventures
  - MAASTRO Innovations B.V. (CSO)

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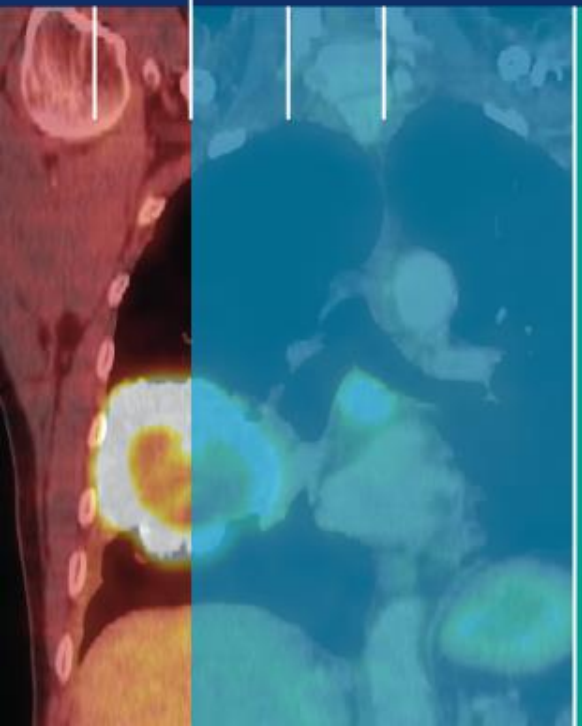
Contents

- Why did we start the CAT* project?
- Barriers to sharing data
- What are we doing with Semantic Web
- A note on Big Data
- Pilot project in Liverpool, NSW

- *CAT=euroCAT, duCAT, VATE, chinaCAT, SeDI etc.

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Why we started the CAT project



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Can we predict a tulip's color by looking at the bulb?







http://www.amystewart.com

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Predicting the tulip color



AUC



1.00



0.72



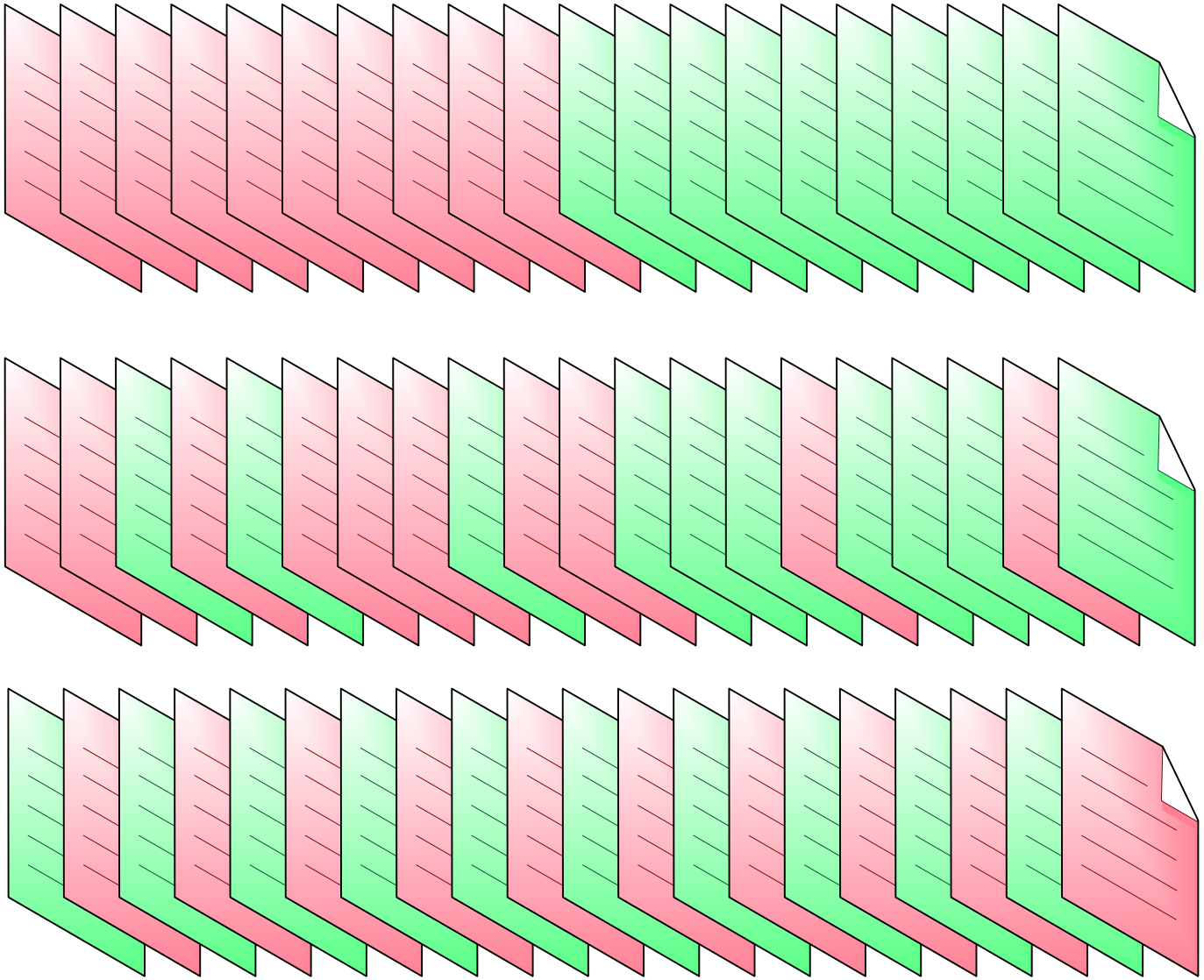
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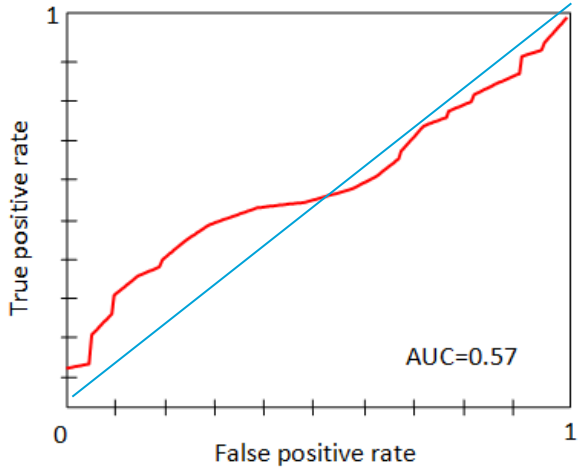
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Experiment



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Prediction by MDs?



NSCLC

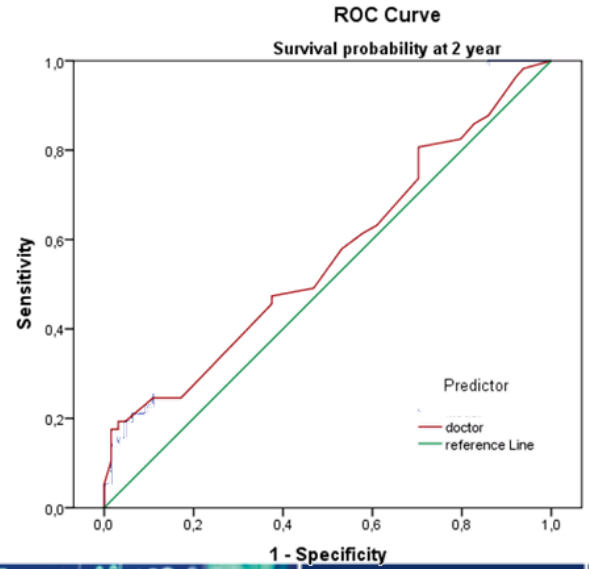
2 year survival

30 patients

8 MDs

Retrospective

AUC: 0.57



NSCLC

2 year survival

158 patients

5 MDs

Prospective

AUC: 0.56

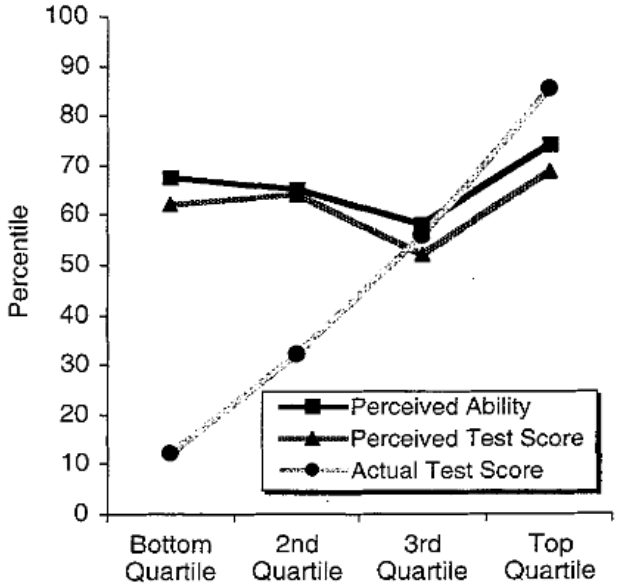


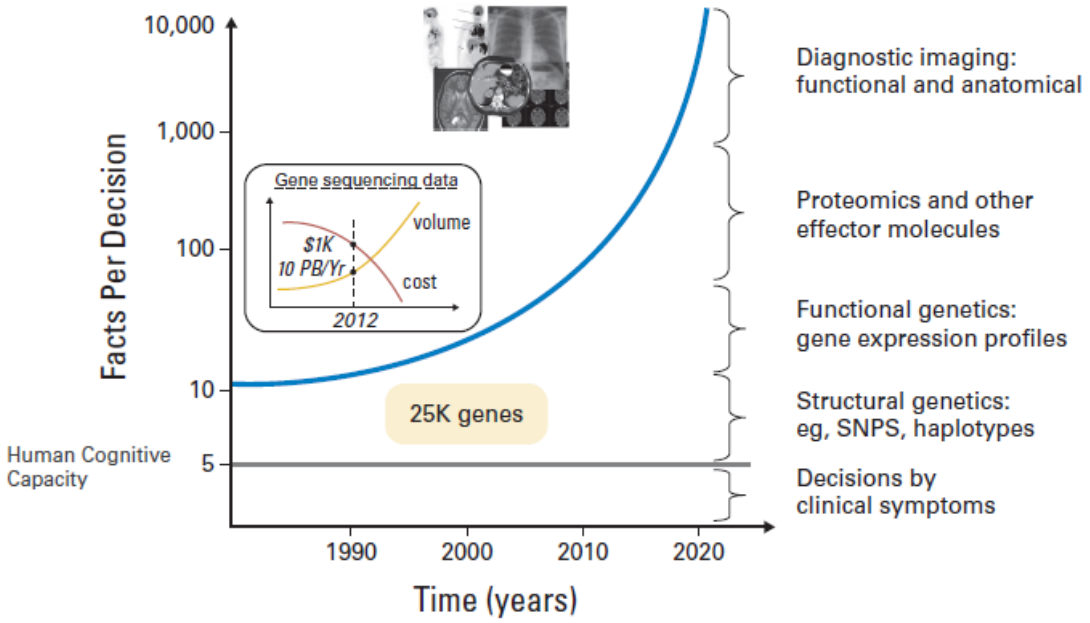
Figure 2. Perceived logical reasoning ability and test performance as a function of actual test performance (Study 2).

Unskilled and unaware of it: *How difficulties in recognizing one's own incompetence leads to inflated self-assessments.* J Pers Soc Psych

Oberije et al.

Kruger et al. 1999

The doctor is drowning



- Explosion of data
- Explosion of decisions
- Explosion of 'evidence'*
  - 3 % in trials, bias
  - Sharp knife

*2010: 1574 & 1354 articles on lung cancer & radiotherapy = 7.5 per day

Half-life of knowledge estimated at 7 years (in young students)

J Clin Oncol 2010;28:4268
JMI 2012 Friedman, Rigby

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Why?

- In radiotherapy we treat patients using technology
- Getting RCT evidence on technology is hard
- Already we can't predict the outcome of RT patients
- This will become worse as our knowledge of cancer and technology increases
- If we can't predict outcome, the treatment decisions will be suboptimal

Future radiotherapy practice will be based on evidence from retrospective interrogation of linked clinical data sources rather than prospective randomized controlled clinical trials

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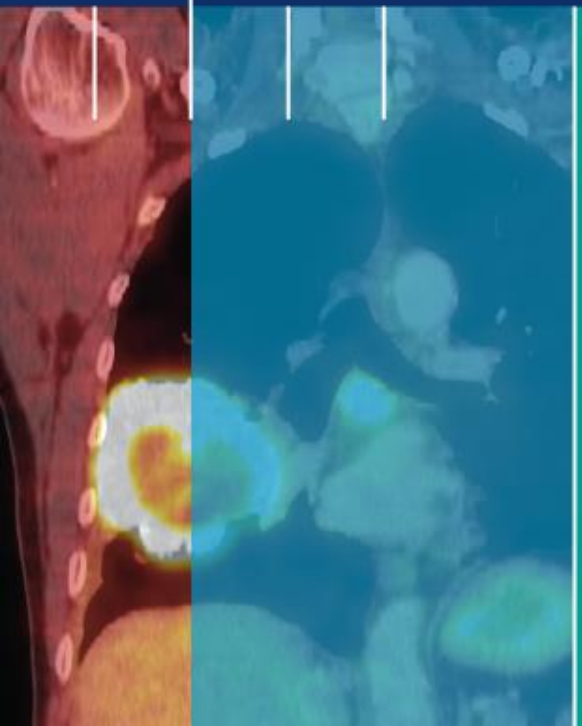
Martin A. Ebert, Ph.D. and Colin G. Orton, Ph.D., Co-Moderators

(Received 13 October 2013; accepted for publication 14 October 2013; published 4 February 2014)

[http://dx.doi.org/10.1118/1.4832139]

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CAT is about improving medical decisions via the creation of an alternative evidence base: **Your Clinic**



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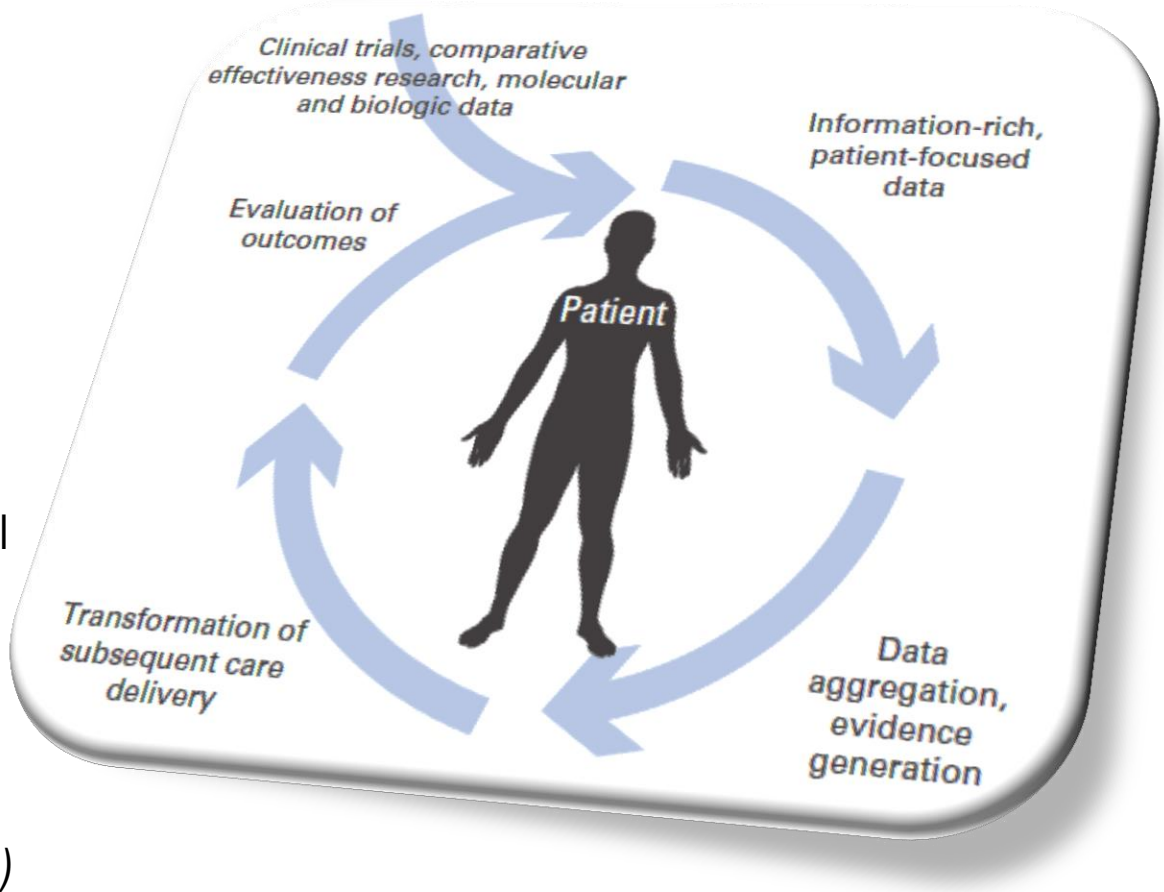
Rapid Learning Health Care

In [..] rapid-learning [..] data routinely generated through **patient care and clinical research** feed into an ever-growing [..] set of **coordinated databases**.

J Clin Oncol 2010;28:4268

[..] rapid learning [..] where we can **learn from each patient** to guide practice, is [..] crucial to guide rational health policy and to contain costs [..].

Lancet Oncol 2011;12:933



Examples:

Radiotherapy CAT (www.eurocat.info)

ASCO's CancerLinQ

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CAT ~ 2005 = MAASTRO Knowledge Engineering

Build **Decision Support Systems** to

individualize patient care
by using **machine learning**

to extract **multifactorial**
personalized prediction models

Theme 2
Learning
(1%)

from **existing databases**
containing **all data on all patients**

that are validated in **external datasets**

Theme 1
Data
(99%)

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Classic linked data solution

- Sharing standardized, highly curated multi-domain data from clinical research programs
  - Via supplemental material / publications
  - Open-data
  - Via the Semantic Web

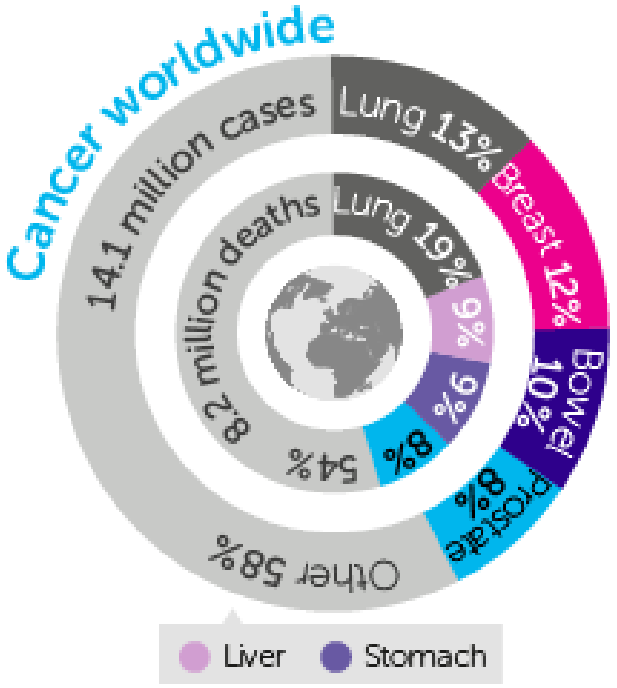
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Back of the envelope

Link clinical radiotherapy treatment planning CT scans for Stage I-IIIB NSCLC to survival

- 10 years incidence 141M Cancer
- 18M Lung cancer
- 14M NSCLC
- 7M Stage I-IIIB
- 3.5M Stage I-IIIB RT

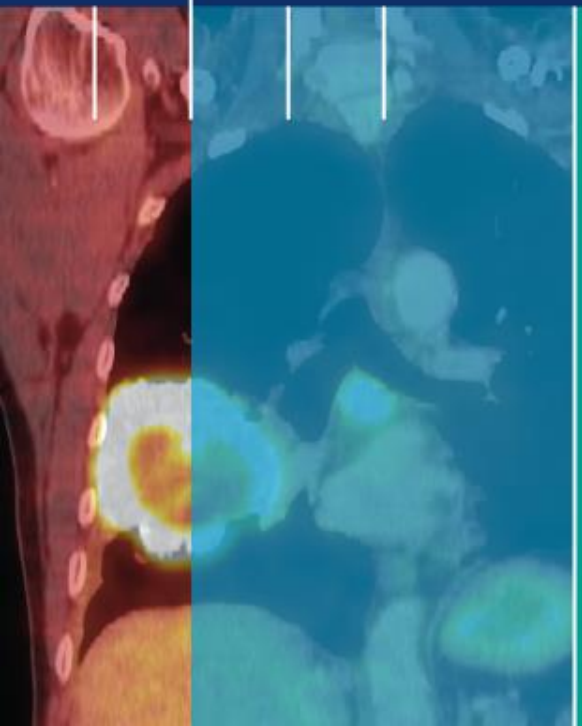
How can we get these 3.5M patients?



http://www.cancerresearchuk.org

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Barriers to sharing data and a way to overcome these



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Barriers to sharing data

[..] the problem is not really technical […]. Rather, the problems are **ethical, political, and administrative**.
Lancet Oncol 2011;12:933

1. Administrative (time to capture, time to curate)
2. Political (value, authorship)
3. Ethical (privacy)

4. Technical



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CAT approach

If sharing is the problem: Don't share the data

If you can't bring the data to the learning application
You have to bring the learning application to the data

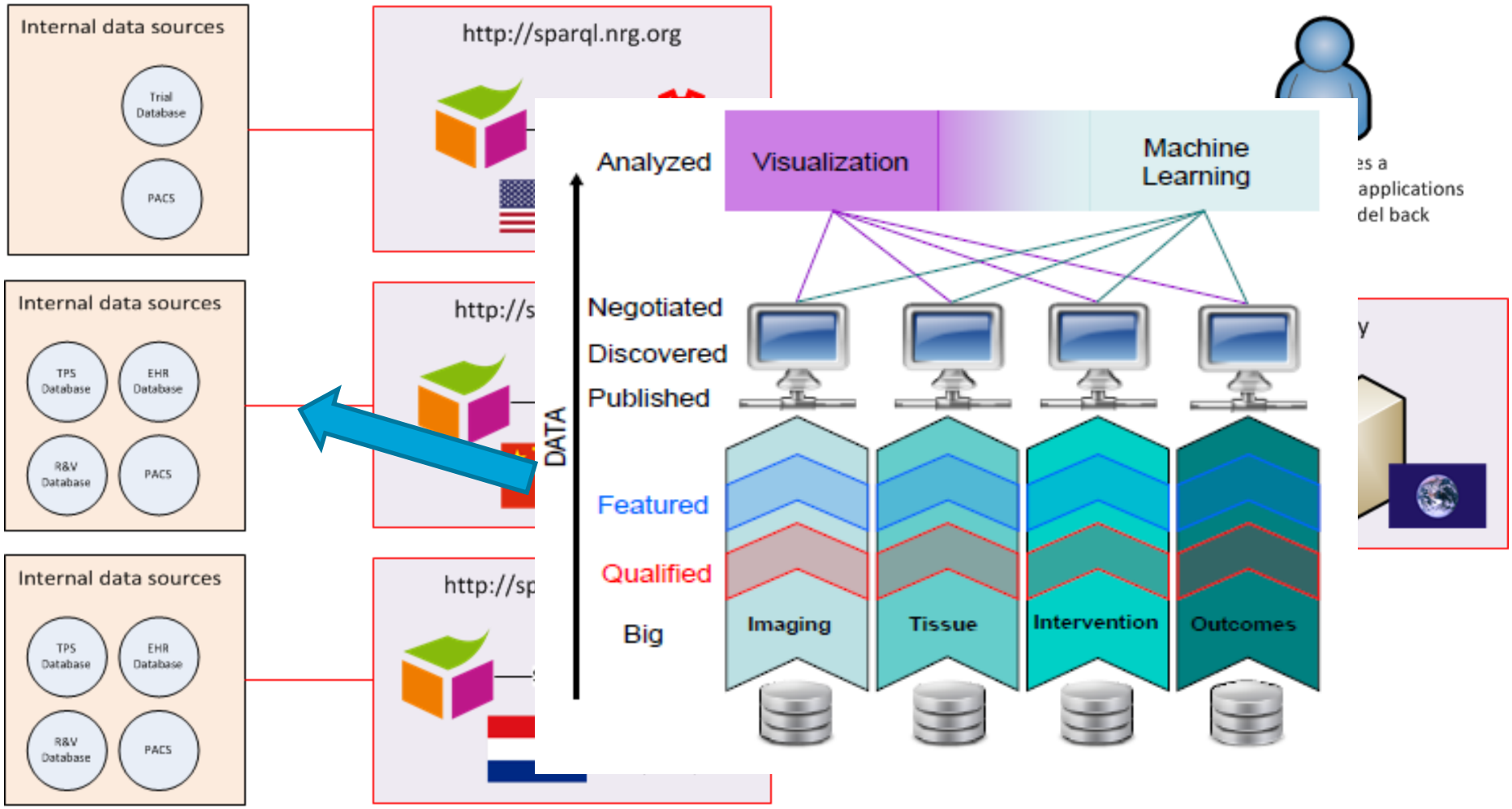
Consequences

- The learning application has to be distributed
- The data has to be readable by an application (i.e. not a human)

- Solution: Sharing standardized highly curated research data
- Solution: Not-sharing non-standardized non-curated clinical data

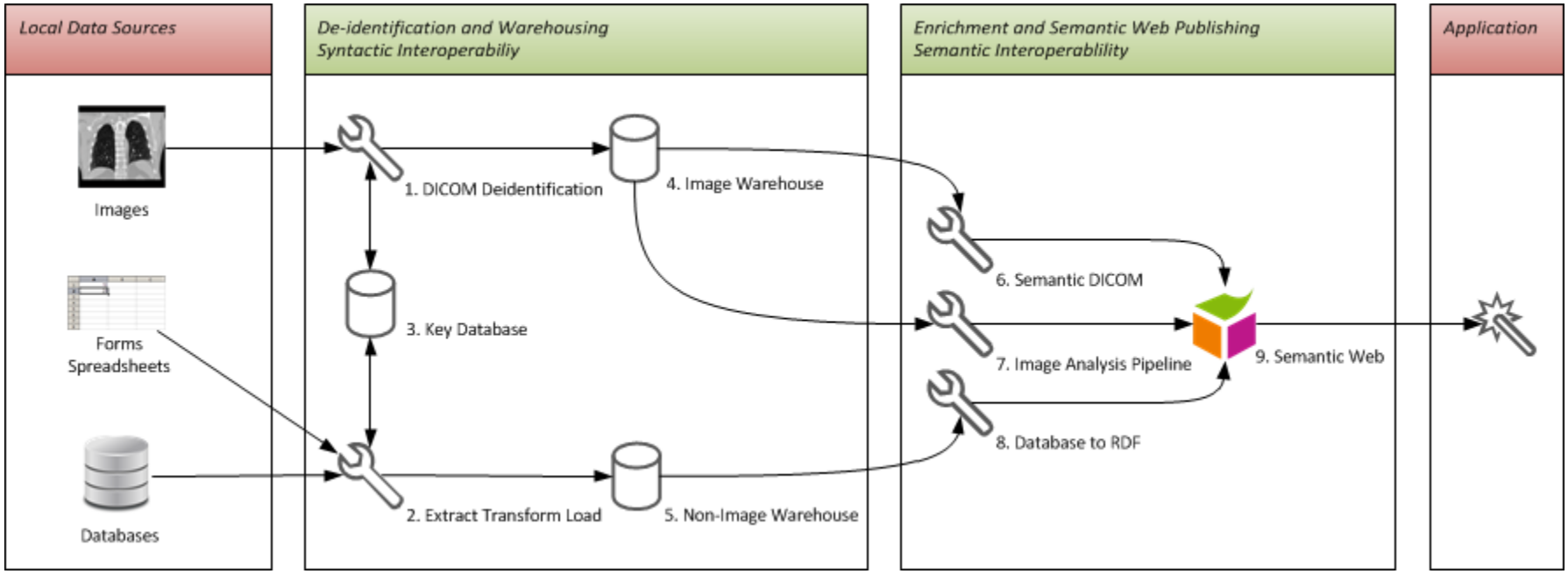
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Under the hood – Publish on the Semantic Web



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In-hospital infra & de-identification

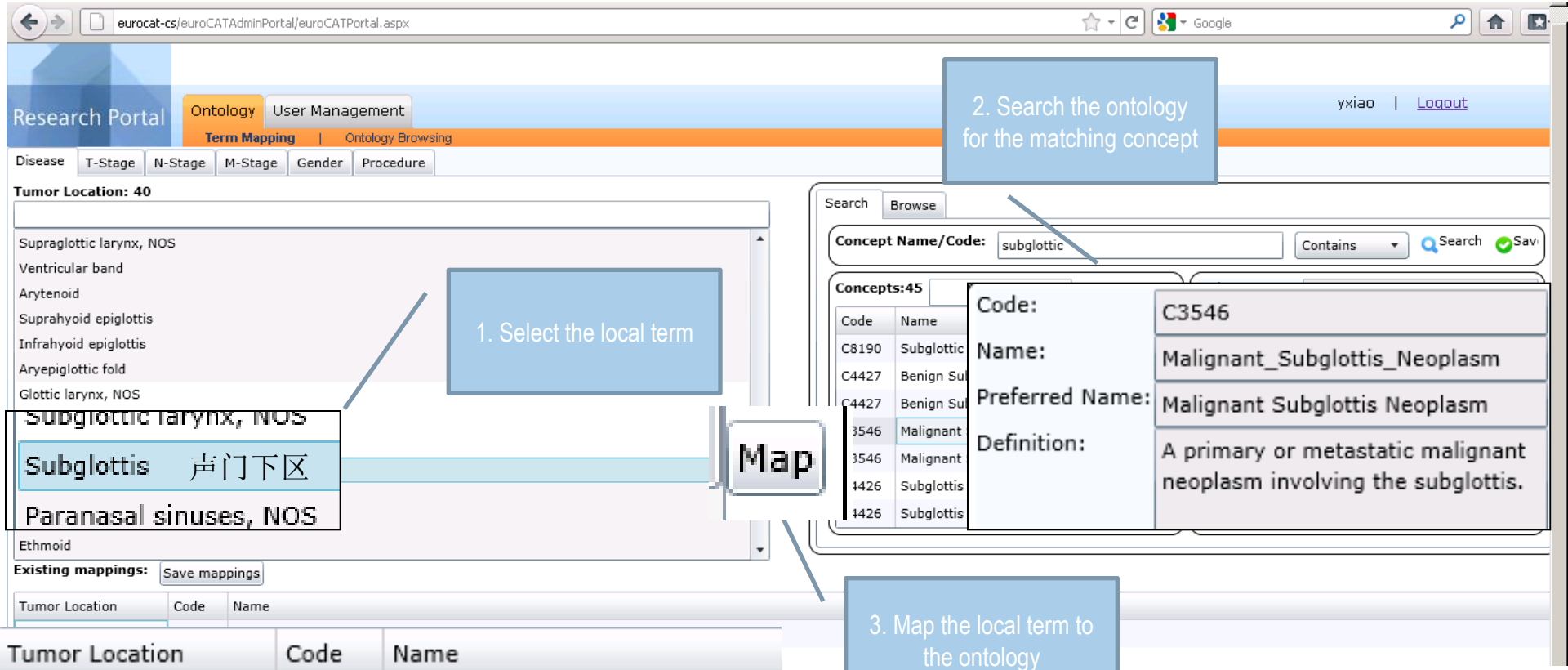


Deidentification:

- Removal of obvious patient identifiers (name, MRN, social security number, email etc.)
- Assign a persistent token pseudonym
- Change (data banding) of obvious but required patient identifiers (everyone born and died on the 15th of the month, part of the postal code)
- No individual patient data leaves the hospital

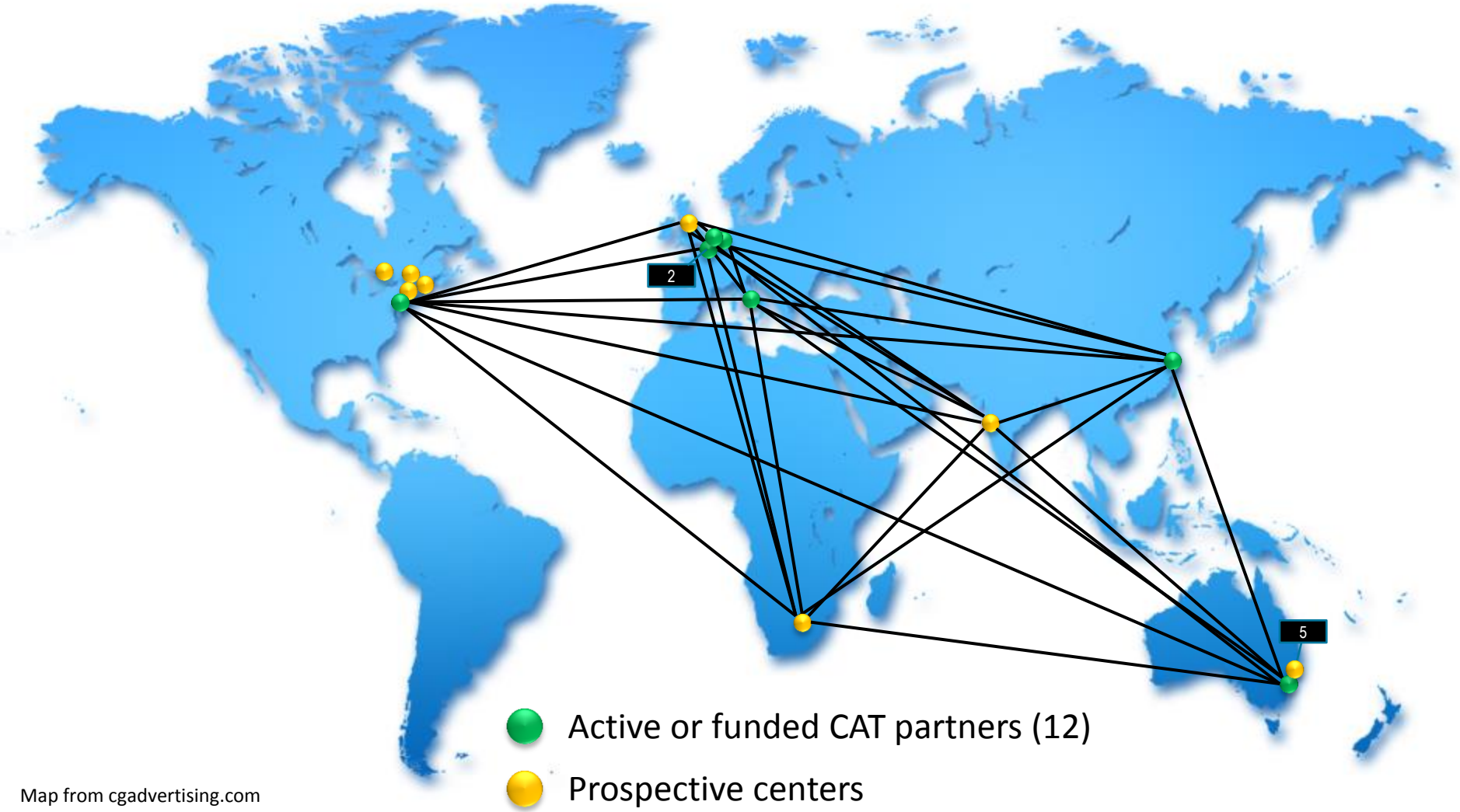
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Ontology – International Coding System



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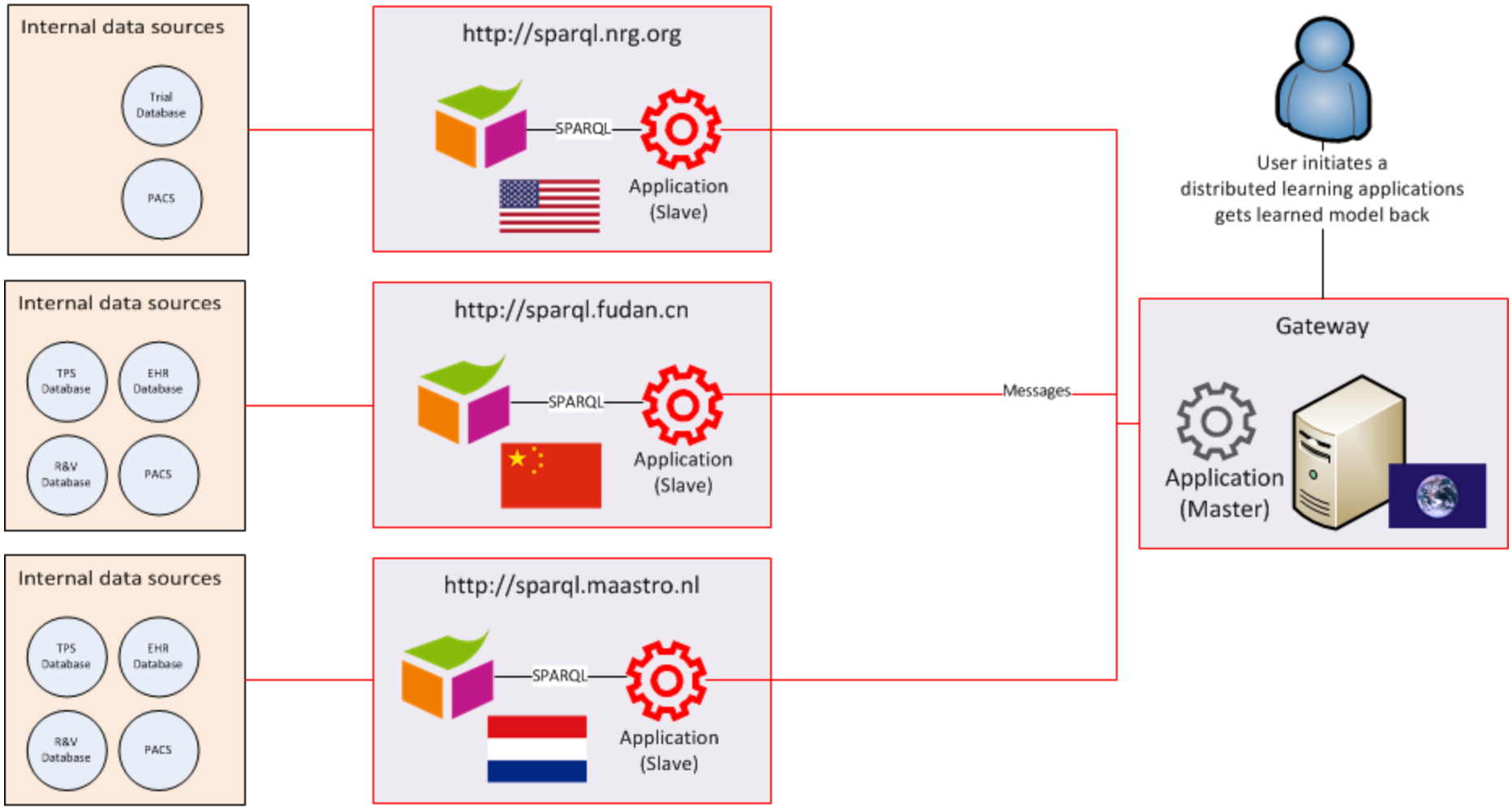
Funded: euroCAT,duCAT,chinaCAT, VATE
New: ozCAT, ukCAT, indiaCAT, “Big Machine” (Can.)



Map from cgadvertising.com

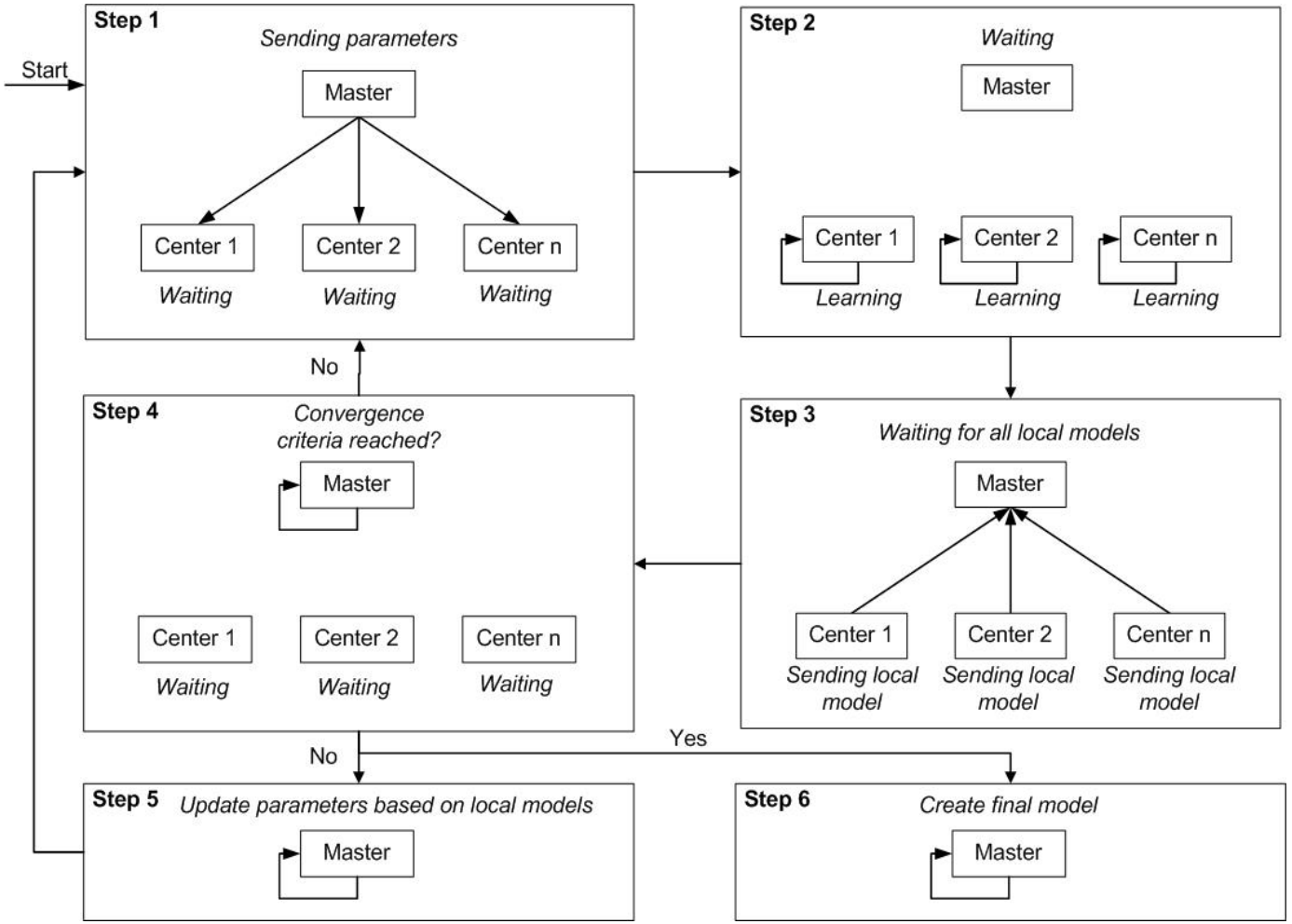
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High level Infra



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Distributed learning



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Does all of that work ? euroCAT's example

- Distributed learning = Centralized learning
- Distributed learning better than learning on individual data

Learn in	Validate in	AUC
Aachen (n=7)	Liège (n=186)	0.61
Eindhoven (n=32)	Liège (n=186)	0.72
Hasselt (n=45)	Liège (n=186)	0.68
Maastricht (n=52)	Liège (n=186)	0.75
Alle 4 samen (n=136)	Liège (n=186)	**0.77**
Alle 5 samen (n=322)	World (n=*inf*)	?

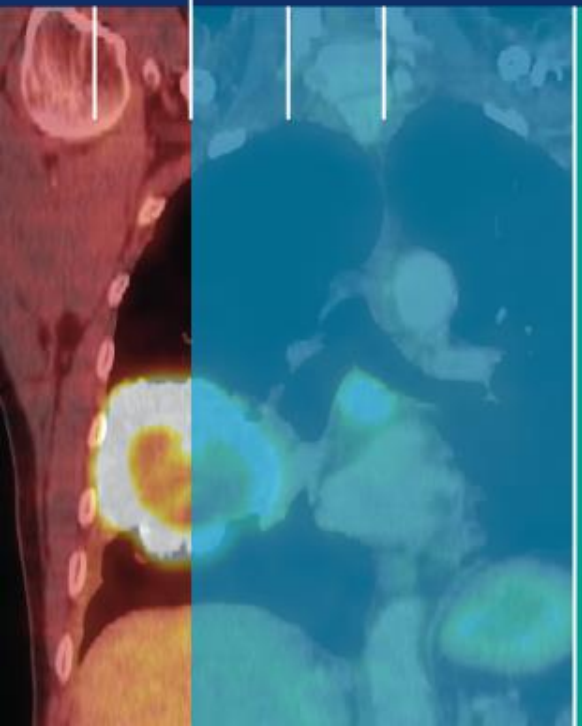
- 550 iterations, two hours (centralized < 1 min)

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ESTRO Forum 2013
R&O Oberije May 2014



A note on Big Data



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Knowledge Engineering Head

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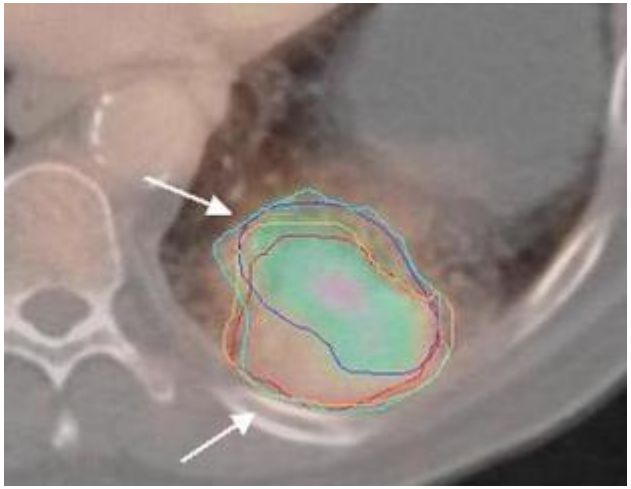
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Big Data in Medicine

Many people would have you believe this is due to genomics, but…

- Volume: “By 2015, the average hospital will have two-thirds of a petabyte of patient data, **80% of which will be unstructured image data** like CT scans and X-rays”.
- Velocity: “Medical Imaging archives are increasing by 20%-40% each year”
- Variety: 2D, 3D and 4D, modalities, contrast agents, multiple time points
- Veracity: Data quality and interpretation/annotation



The size of the genomics data from clinical patients at MAASTRO: 0 bytes.

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Semantic DICOM?

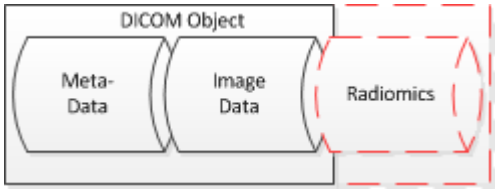
- DICOM is the standard for medical imaging
- Digital Imaging and Communications in Medicine
- Paper standard, 20 parts and thousands of pages
- Syntactic standard
- A blog (“So you think HTML is hard? Try DICOM!”)
  - The standard is long.
  - The standard is old.
  - The standard is ambiguous.
  - The standard is confusing.
  - Major vendors don’t comply to the standard.

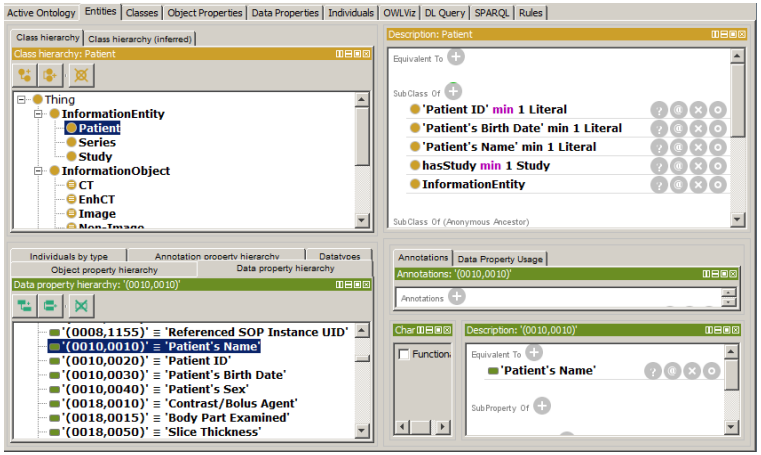
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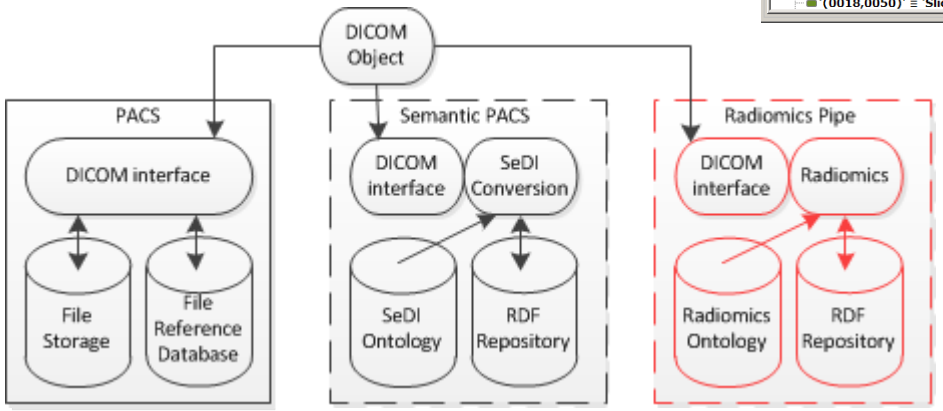
Publishing image data as linked data

eurostars™

SOHARD SOFTWARE







SeDI

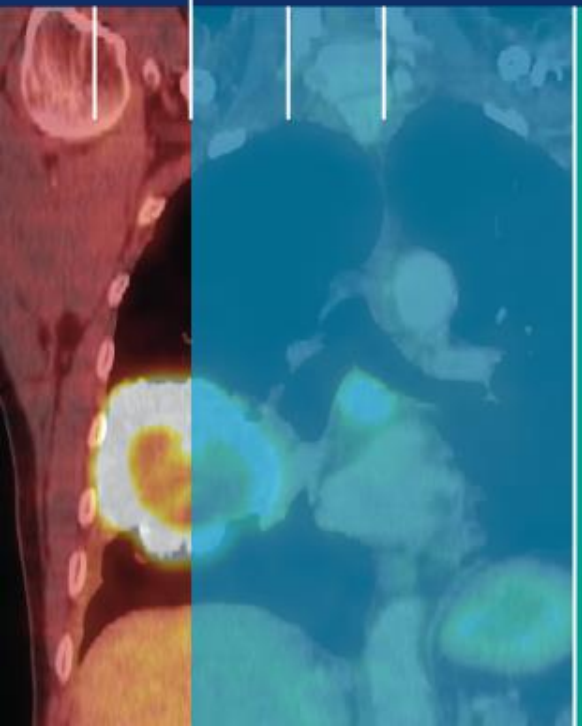


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And how does all this benefit the patient? An Australian Pilot



Andre Dekker, PhD

Medical Physicist

Information & Services Head

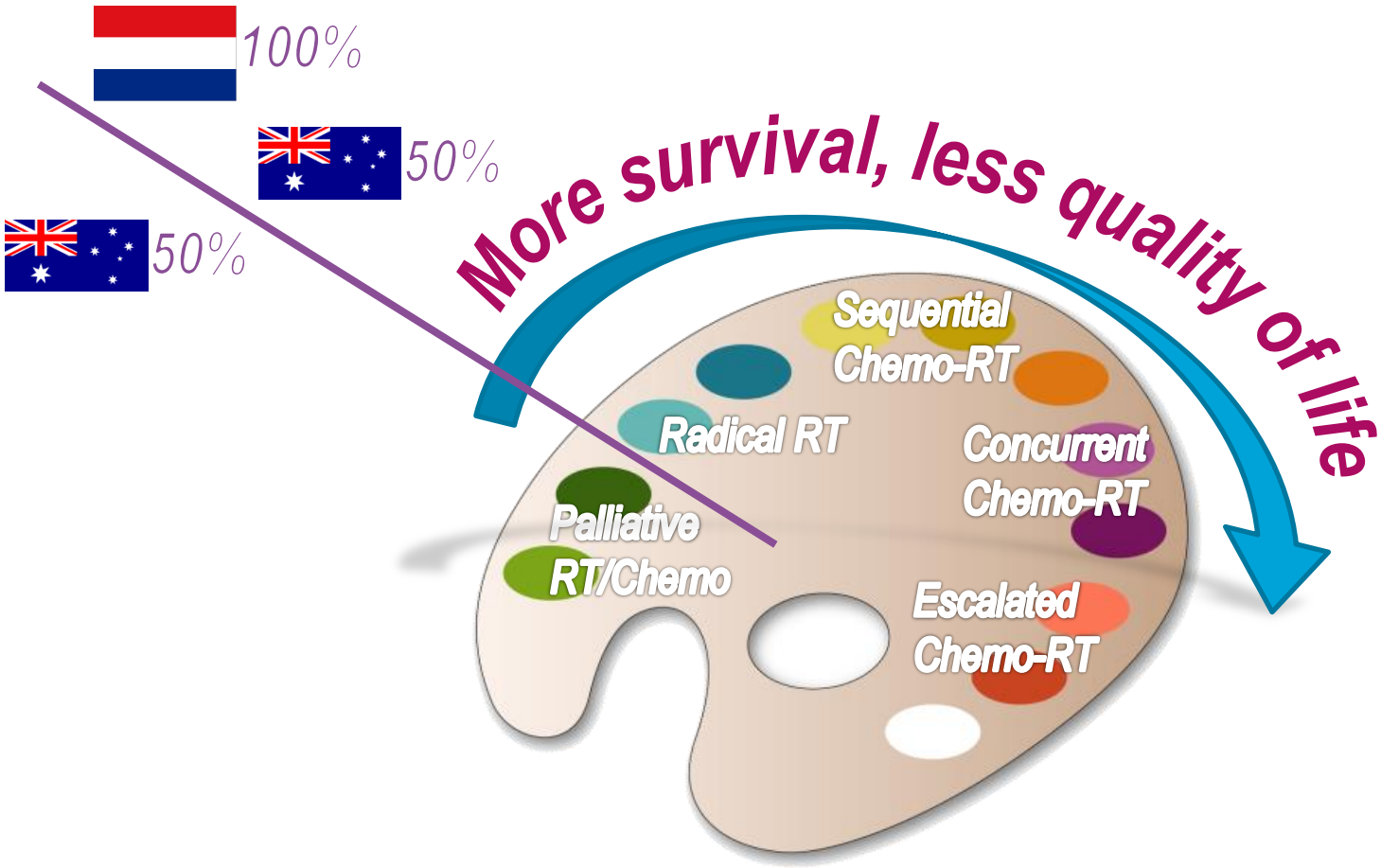
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Guideline vs. the real world in lung cancer

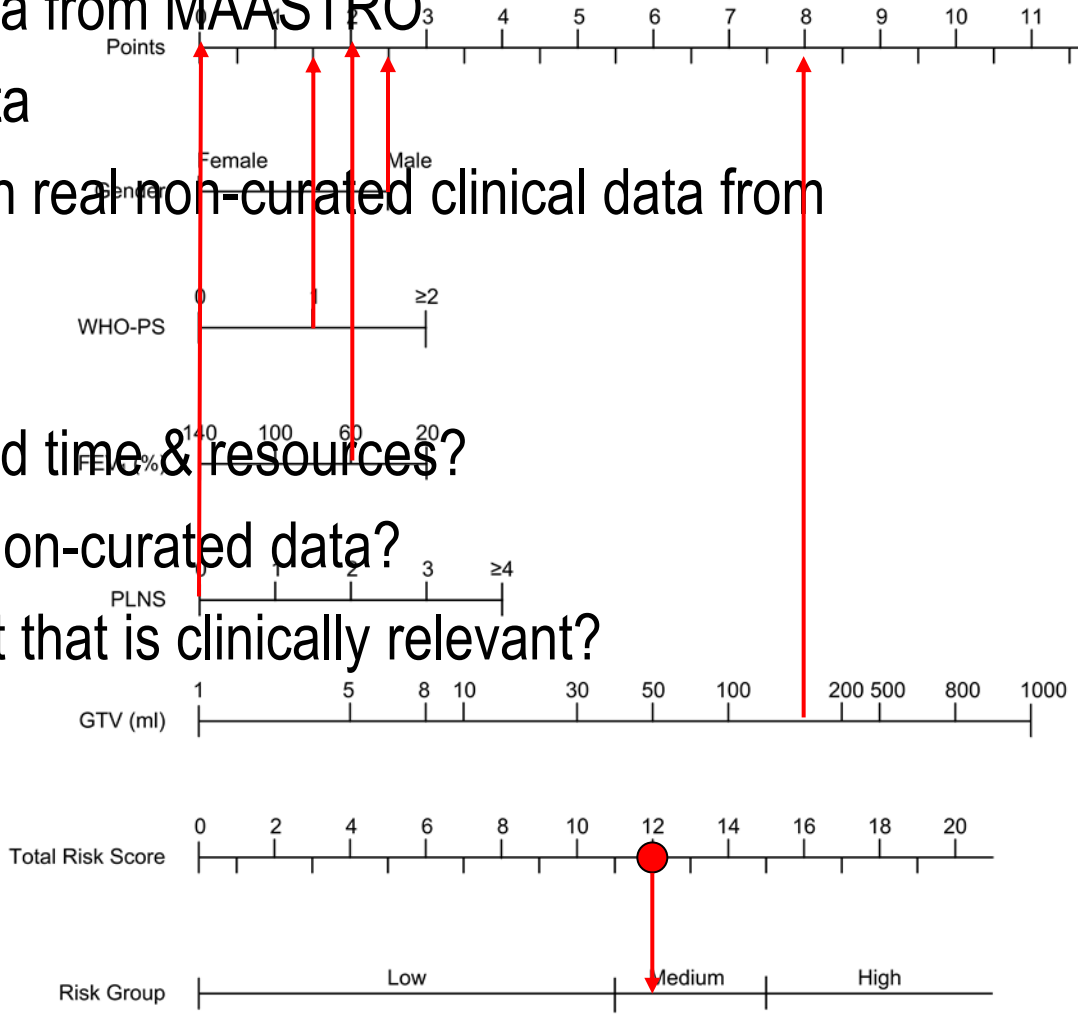


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Liverpool and Macarthur CC project

- DSS: Nomogram for OS for curatively treated NSCLC (AUC 0.75)
- Learned from curated data from MAASTRO
- Rapid Learning = real data
- "Commission" the DSS on real non-curated clinical data from Liverpool and Macarthur

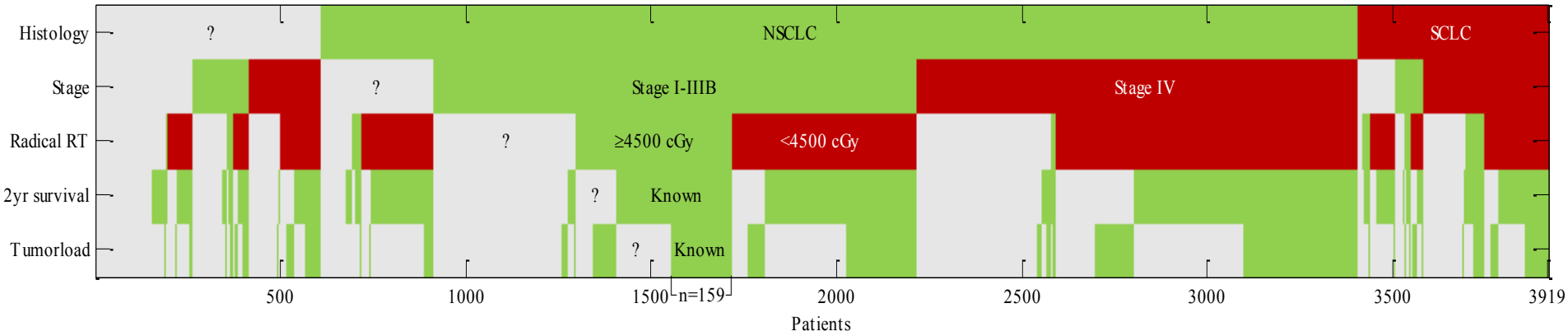
- Is this possible with limited time & resources?
- Does the model work in non-curated data?
- What can we learn from it that is clinically relevant?



Dehing-Oberije (MAASTRO), IJROBP 2009;74:355

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Patients found meeting inclusion criteria

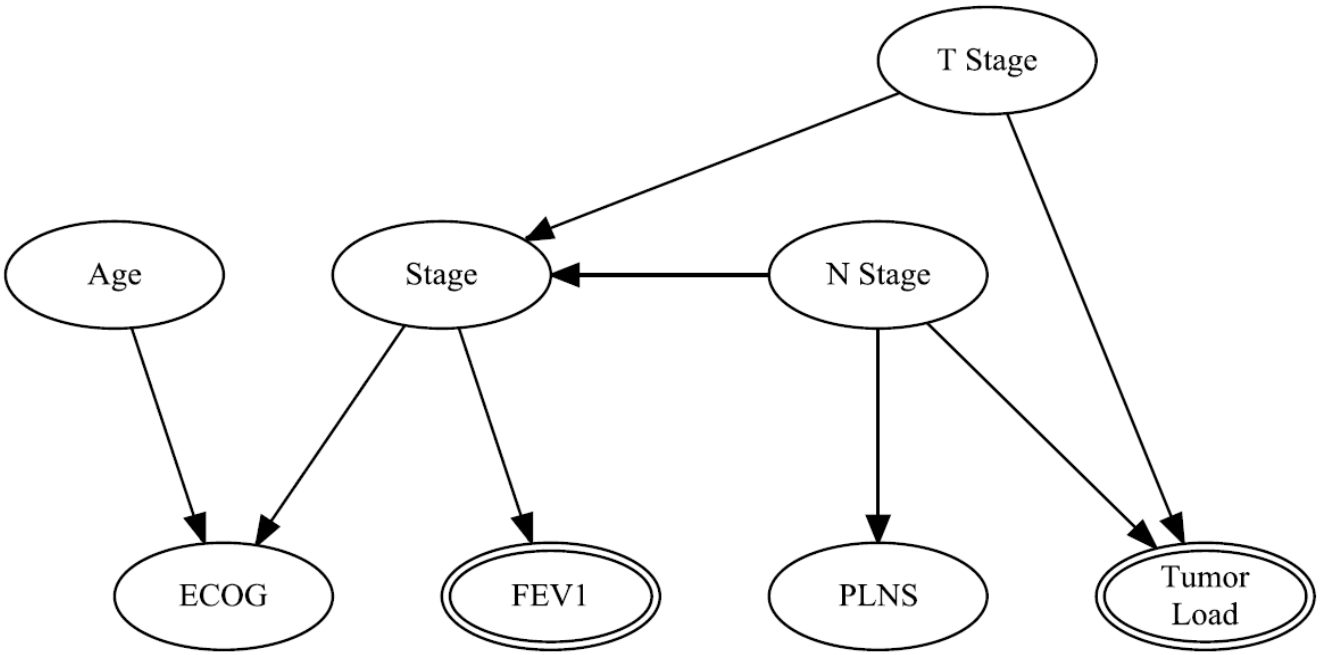


- Practice insight: 50%+ of stage I-IIIB treated with non-radical treatments
- Sobering: 159 eligible from 3919 patients

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Missing data in included patients

		Training cohort	Commissioning cohort	
Site		MAASTRO	Liverpool	
# Patients		322	159	
Age (years)	Mean±			
Gender	Male			
	Femal			
Stage	I			
	II			
	IIIA			
	IIIB			
	Missin			
ECOG	0			
	1			
	≥2	52(16%)	13(8%)	
	Missing	7(2%)	25(16%)	

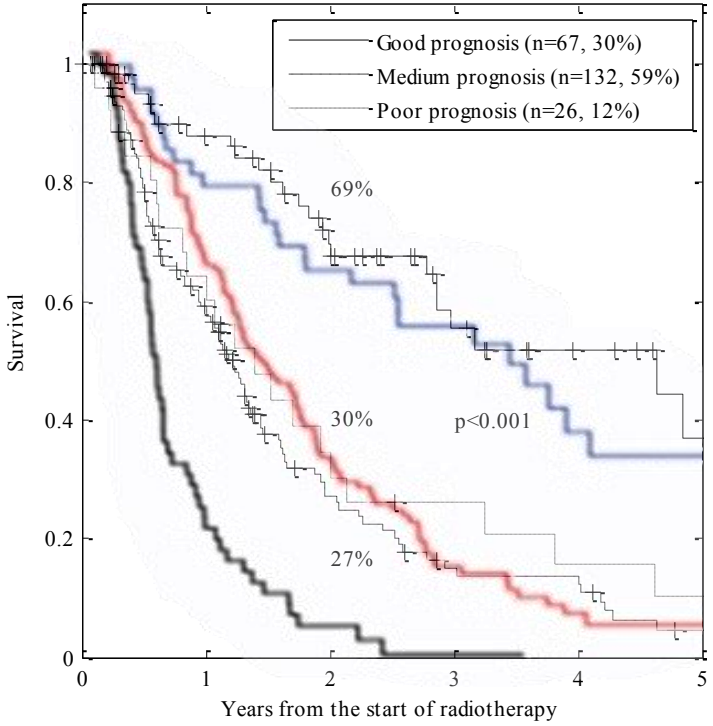


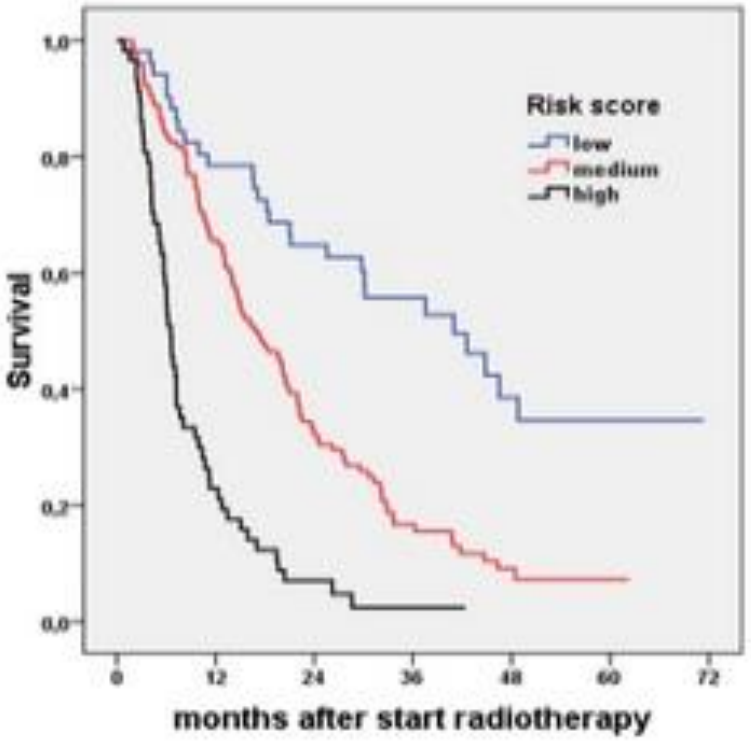
		Training cohort	Commissioning cohort	
Site		MAASTRO	Liverpool	
# Patients		322	159	
			±20	
			(60%)	
			±147	p<0.001
			*(0%)	
			(0%)	
			(0%)	
			(0%)	
			(0%)	
			(0%)	
			*(100%)	
			(36%)	
	No	219(68%)	101(64%)	
	Missing	0(0%)	0*(0%)	

Still a lot of missing data -> imputation necessary (Bayesian Network)

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Results





DSS works, but only to discriminate between good and medium/poor

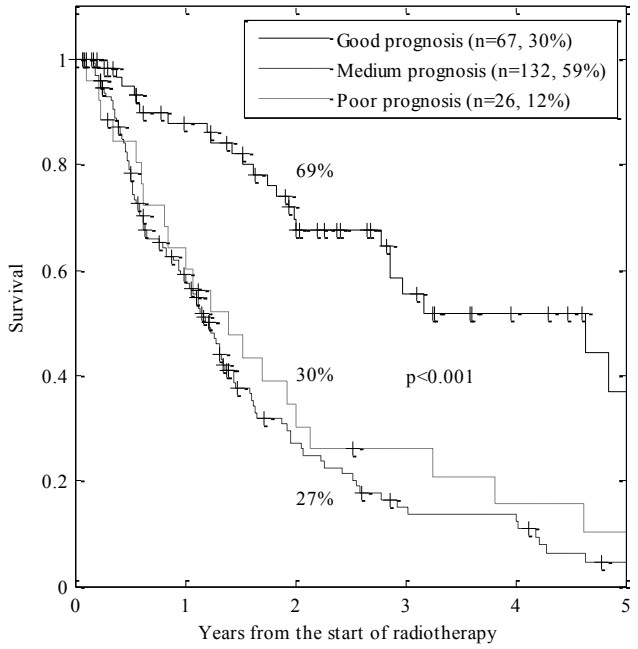
Better than TNM stage

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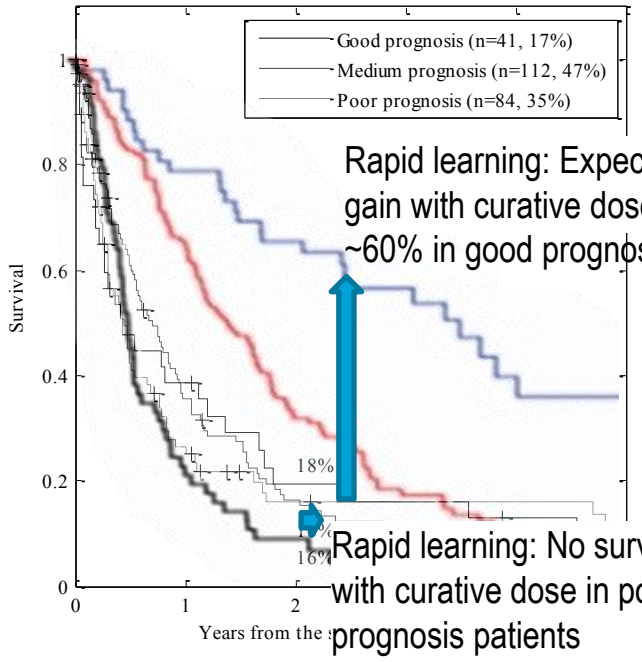
What did Liverpool learn? What did MAASTRO learn?

- Rethink palliative treatments in good prognosis patients
- Rethink curative treatments in poor prognosis patients

radical RT treatments



palliative RT treatments



Rapid learning: Expected survival gain with curative dose from 18 to ~60% in good prognosis patients

Rapid learning: No survival gain with curative dose in poor prognosis patients

routine data, realistic quality, good evidence?

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Conclusion

- Link to youtube movie
- Semantic Web technology is helping us to open up clinics for rapid learning

Working on now

- Shape expressions
- Secure SPARQL
- Ontology extensions (esp. predicates) and integrations
- Cataloguing
- Physician facing interfaces for querying and analysis
- Probabilistic predicates – Bayesian Networks

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Models built & validated

Lung cancer

- Survival
- Lung dyspnea
- Lung dysphagia

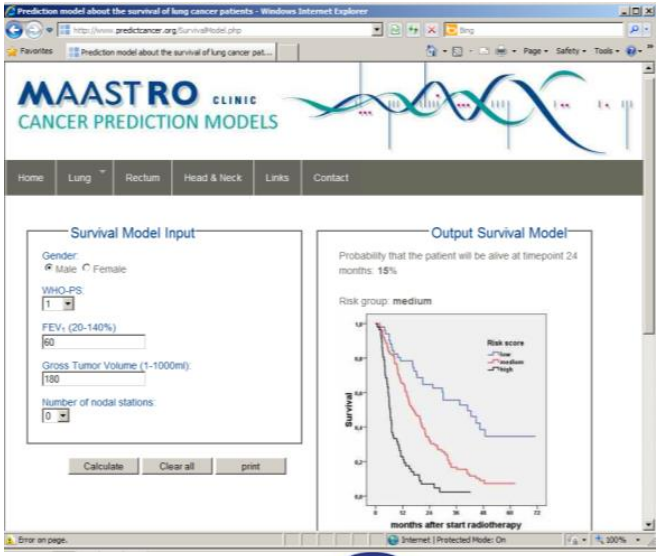
H&N cancer

- Local recurrences
- Overall survival
- Cost Effectiveness IMPT vs. IMRT

www.predictcancer.org

Rectal cancer

- Tumor response
- Local recurrences
- Distant metastases
- Overall survival



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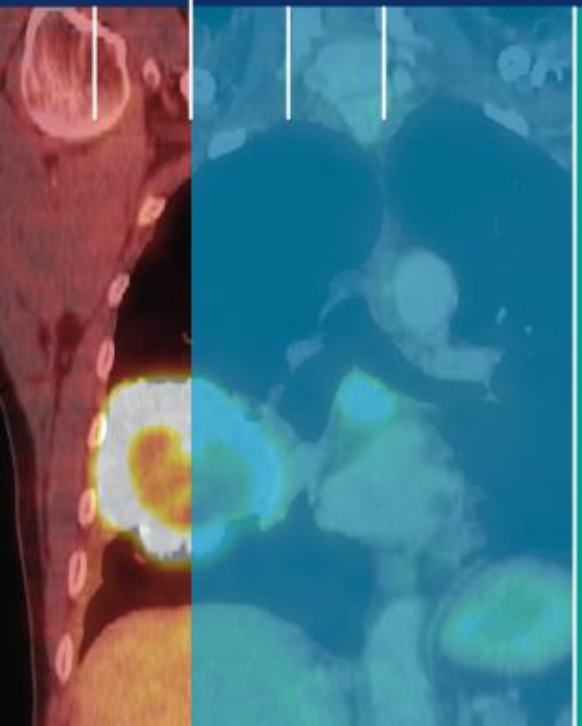
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Met de steun van het Europees Fonds voor Regionale Ontwikkeling.

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